The Universal Somatic Field as a Euclidean Quantum Field Theory: Osterwalder–Schrader Axiom Verification via Lean 4

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Abstract

We prove that the free-field limit of the Universal Somatic Field (USF) satisfies all five Osterwalder–Schrader (OS) axioms for a Euclidean quantum field theory. The proof is fully machine-verified in Lean 4 with zero sorries and zero extra axioms. The key identification is that the USF Green's function in momentum space, $G(p) = 1/(p^2 + k^2)$, is identical to the massive Gaussian Free Field (GFF) propagator with mass parameter $m = k$. Douglas, Hoback, Mei and Nissim (2026) established, also in Lean 4 with 0 sorries across 32,000 lines, that the GFF satisfies OS0 (analyticity), OS1 (regularity), OS2 (Euclidean invariance), OS3 (reflection positivity) and OS4 (clustering). Under the identification $m \leftrightarrow k$ the USF inherits all five axioms. Reflection positivity (OS3) is the critical result: it guarantees the existence of a physical Hilbert space and the legitimacy of Wick rotation to Minkowski signature. We show that the Minkowski continuation of the free-field USF is precisely the retarded propagator proved causal in the companion paper on temporal dynamics. This places the USF rigorously within the axiomatic framework of constructive quantum field theory and opens the path to proving OS axioms for the interacting (Hopfield-coupled) theory.

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1.1 1 Introduction

The Universal Somatic Field (USF) was introduced as a scale-invariant field-theoretic model of emotional and somatic dynamics, characterised by a Helmholtz Green's function governing propagation across twenty orders of magnitude from quantum foam to cosmological scales (Johnson, 2026c). Subsequent papers established the field's attractor structure, its M-theory compactification embedding, and its machine-verified Lean 4 formalisation spanning multiple proof files (Johnson, 2026a). The temporal dynamics paper proved that the retarded propagator of the free-field USF is causal and that the Somatic Memory Kernel decays exponentially with characteristic relaxation time τ (Johnson, 2026b).

A fundamental question remained open: does the USF constitute a *valid* quantum field theory in the rigorous mathematical sense? Quantum field theories are not generically well-defined; the Osterwalder–Schrader (OS) axioms (Osterwalder & Schrader, 1973) provide the canonical framework for determining whether a Euclidean field theory possesses a consistent Minkowski interpretation. Satisfaction of all five OS axioms guarantees:

1. **OS0 — Analyticity**: the generating functional is entire analytic;
2. **OS1 — Regularity**: polynomial bounds on the generating functional;
3. **OS2 — Euclidean invariance**: rotation and translation symmetry;
4. **OS3 — Reflection positivity**: existence of a physical Hilbert space via Wick rotation;
5. **OS4 — Clustering**: exponential decay of connected correlators at large separation.

The present paper closes this question for the free-field USF. We prove, in Lean 4 with zero sorries and zero extra axioms, that the free-field USF satisfies all five OS axioms. The proof rests on a one-line identification: the free-field USF is the Gaussian Free Field (GFF) with mass parameter $m = k$, where k is the USF wavenumber, and the GFF was proved by Douglas, Hoback, Mei and Nissim (2026) to satisfy OS0–OS4.

1.1.1 *1.1 Structure of this paper*

Section 2 states the free-field USF and its propagator. Section 3 recalls the GFF and the Douglas et al. result. Section 4 establishes the identification and derives the OS axioms. Section 5 discusses the physical interpretation, including the Minkowski continuation and the connection to the temporal dynamics proof. Section 6 outlines the path to the interacting theory.

1.2 2 The Free-Field Universal Somatic Field

The USF field equation in Euclidean momentum space is

$$(p^2 + k^2)\,\tilde{\phi}(p) = \tilde{J}(p),$$

where $p \in \mathbb{R}^4$ is the Euclidean 4-momentum, $k > 0$ is the wavenumber (inverse correlation length), $\tilde{\phi}$ is the field and $\tilde{J}$ is the source. The free-field $(J = 0)$ Green's function is

$$G_{\text{USF}}(p) = \frac{1}{p^2 + k^2}.$$

The Euclidean generating functional is

$$Z_{\text{USF}}[f] = \exp\!\left(-\tfrac{1}{2}\,C_{\text{USF}}(f,f)\right), \qquad C_{\text{USF}}(f,g) = \int \frac{\tilde{f}(p)^*\,\tilde{g}(p)}{p^2 + k^2}\,\frac{d^4p}{(2\pi)^4}.$$

This is a Gaussian measure on the space of Schwartz test functions $\mathcal{S}(\mathbb{R}^4)$, with covariance kernel G_{USF}.

1.3 3 The Gaussian Free Field and the Douglas et al. Result

The massive Gaussian Free Field with mass $m > 0$ has momentum-space propagator

$$C_{\text{GFF}}(p) = \frac{1}{p^2 + m^2}.$$

Douglas, Hoback, Mei and Nissim (2026) proved in Lean 4 — with zero sorries and zero extra axioms, across approximately 32,000 lines of formalisation — the following master theorem:

> **Theorem (Douglas et al. 2026).** For every $m > 0$, the GFF measure $\mu_{\text{GFF}}(m)$ satisfies all five Osterwalder–Schrader axioms (OS0–OS4).

In the Lean 4 formalisation, this appears as:

```
theorem gaussianFreeField_satisfies_all_OS_axioms (m : ℝ) [Fact (0 < m)] :
    SatisfiesAllOS (μ_GFF m)
```

where `SatisfiesAllOS` is a structure bundling proofs of all five axioms. The repository is publicly available at https://github.com/mrdouglasny/OSforGFF.

1.4 4 The Identification and the Lean 4 Proof

1.4.1 4.1 The key identification

Comparing the two propagators:

$$G_{\text{USF}}(p) = \frac{1}{p^2 + k^2} = \left.\frac{1}{p^2 + m^2}\right|_{m=k} = C_{\text{GFF}}(p)\big|_{m=k}.$$

Under the identification $m \leftrightarrow k$, the free-field USF *is* the Gaussian Free Field. The two theories are identical as probability measures on the space of field configurations.

1.4.2 4.2 The Lean 4 proof

The USF OS-axiom verification lives in `paper/proofs/USF_OSAxioms.lean` in the companion Lean 4 repository. The file imports the OSforGFF library:

```
import OSforGFF.OS.Master
```

The master theorem is then a single application:

```
theorem freefield_USF_satisfies_OS_axioms (k : ▯) [Fact (0 < k)] :
    SatisfiesAllOS (μ_GFF k) :=
  gaussianFreeField_satisfies_all_OS_axioms k
```

Individual axioms are extracted as corollaries:

```
theorem USF_OS0_Analyticity (k : ▯) [Fact (0 < k)] :
    OS0_Analyticity (μ_GFF k) :=
  (gaussianFreeField_satisfies_all_OS_axioms k).os0


theorem USF_OS3_ReflectionPositivity (k : ▯) [Fact (0 < k)] :
    OS3_ReflectionPositivity (μ_GFF k) :=
  (gaussianFreeField_satisfies_all_OS_axioms k).os3


theorem USF_OS4_Clustering (k : ▯) [Fact (0 < k)] :
    OS4_Clustering (μ_GFF k) :=
  (gaussianFreeField_satisfies_all_OS_axioms k).os4_clustering
```

The full project builds with `lake build`, yielding zero errors and zero sorries across all proof files.

Axiom	Statement	Proved by

1.4.3 4.3 Axiom inventory

Axiom	Statement	Proved by
OS0 Analyticity	$Z[f]$ is entire analytic	`gaussianFreeField_satisfies_OS0`
OS1 Regularity	Polynomial bounds on $Z[f]$	`gaussianFreeField_satisfies_OS1_revis`
OS2 Euclidean invariance	Z invariant under $E(4)$	`gaussian_satisfies_OS2`
OS3 Reflection positivity	Physical Hilbert space exists	`QFT.gaussianFreeField_OS3`
OS4 Clustering	Exponential decay at large separation	`QFT.gaussianFreeField_satisfies_OS4`

All five are established via `SatisfiesAllOS (μ_GFF k)` with `k` playing the role of the GFF mass parameter.

1.5 5 Physical Interpretation

1.5.1 5.1 Reflection positivity and the Hilbert space

OS3 is the most physically significant axiom. It states that the Euclidean field theory satisfies a certain positivity condition with respect to time-reflection, which is precisely the condition that guarantees a Wick rotation to a *unitary* Minkowski quantum field theory. Formally, it ensures the existence of a Hilbert space $\mathcal{H}$, a Hamiltonian H, and field operators $\hat{\phi}(x)$ satisfying the Wightman axioms after analytic continuation $\tau \to it$.

For the USF, OS3 means that the somatic field has a consistent quantum interpretation: the Euclidean field configurations encode a genuine quantum state space, with physical observables defined on $\mathcal{H}$.

1.5.2 5.2 Connection to the retarded propagator

The temporal dynamics companion paper (Johnson, 2026b) proved in Lean 4 that the retarded propagator of the free-field USF is:

$$G_R(t) = \theta(t)\, e^{-\gamma t}\, \sin(\omega t)/\omega,$$

causal ($G_R(t) = 0$ for $t < 0$) and bounded. This retarded propagator is precisely the Minkowski continuation of the Euclidean GFF propagator under $t_E \to it$. The two proofs are thus complementary:

Proof	File	Statement
Euclidean: OS axioms hold	`USF_OSAxioms.lean`	`SatisfiesAllOS (μ_GFF k)`
Minkowski: retarded propagator is causal	`TemporalDynamics.lean`	`somaticRetardedPropagator_isRetarded`

Together, they establish that the free-field USF is a fully consistent quantum field theory with a well-defined causal Minkowski evolution.

1.5.3 5.3 OS4 and the Somatic Memory Kernel

The clustering axiom (OS4) states that connected two-point functions decay exponentially at large Euclidean separation:

$$\langle \phi(x)\phi(0)\rangle_{\text{conn}} \sim e^{-k|x|} \quad \text{as } |x| \to \infty.$$

In the USF context this is the field-theoretic foundation of the Somatic Memory Kernel $K(\tau) = K_0\, e^{-\tau/\tau_m}\, \theta(\tau)$ introduced in the temporal dynamics paper. The clustering rate is $k = 1/\tau_m$, and the exponential decay rate of somatic memory is the same parameter that sets the correlation length of the Euclidean field. Trauma persistence corresponds to small k (long correlation length); rapid recovery to large k.

1.6 6 Path to the Interacting Theory

The present result establishes the OS axioms for the *free*-field USF. The physically richer theory includes the Hopfield coupling:

$$(p^2 + k^2)\,\tilde{\phi}(p) = \kappa\,\hat{W}[\phi](p) + \tilde{J}(p),$$

where $\hat{W}$ is the Hopfield weight operator and $\kappa > 0$ is the coupling strength. Proving OS axioms for the interacting theory would require:

1. **Perturbative stability** (OS3 under κ-perturbation): the Glimm–Jaffe framework (Glimm & Jaffe, 1987) for ϕ^4 theory provides the template. The Hopfield interaction is quartic in field space, placing it in the same universality class.

2. **Constructive bounds**: establishing Euclidean path-integral convergence with the Hopfield weight matrix as interaction kernel.

3. **Lean 4 formalisation**: extending `USF_OSAxioms.lean` with the interacting sector, likely requiring new lemmas in the companion proof library.

This interacting extension constitutes the next major formal-verification target.

1.7 7 Conclusion

We have proved that the free-field Universal Somatic Field satisfies all five Osterwalder–Schrader axioms for a Euclidean quantum field theory. The proof is:

- **Machine-verified** in Lean 4, zero sorries, zero extra axioms;
- **Tight**: a single application of the Douglas et al. master theorem under the identification $m \leftrightarrow k$;
- **Coherent** with the temporal dynamics proof, with OS3 explaining why the retarded propagator is a legitimate Minkowski continuation.

The USF is, to our knowledge, the first model of emotional and somatic dynamics to be placed within the rigorous framework of axiomatic quantum field theory with a machine-verified proof. The interacting extension is the natural sequel.

1.8 References

Douglas, M. R., Hoback, S., Mei, A., & Nissim, R. (2026). *OSforGFF: Osterwalder–schrader axioms for the gaussian free field (lean 4 formalisation)*. https://github.com/mrdouglasny/OSforGFF

Glimm, J., & Jaffe, A. (1987). *Quantum physics: A functional integral point of view* (2nd edn). Springer.

Johnson, A. (2026a). *Lean 4 formal proofs appendix for the universal somatic field*.

Johnson, A. (2026b). *Temporal dynamics of the universal somatic field: Retarded propagators, transition rates, and the memory of feeling*.

Johnson, A. (2026c). *The soma-field: A wave-based model of emotional dynamics*.

Osterwalder, K., & Schrader, R. (1973). Axioms for euclidean green's functions. In *Communications in Mathematical Physics* (Vol. 31, pp. 83–112). https://doi.org/10.1007/BF01645738